VISION-LANGUAGE MODELS · LECTURE 7

Text-to-Image Generation

Outline

I **Latent Diffusion** — Fixing the cost of generation via diffusion model

II **Text Conditioning** — Adding control — steer denoising with a prompt via cross-attention

III **The Full Pipeline** — Stable Diffusion and overview of DiT

IV **Evaluation** — Scoring generated output: is it realistic? Does it match the prompt?

PART I

Latent Diffusion

The cost of diffusion in pixel space

How many pixels are generated by standard (DDPM) diffusion?
For a 512 x 512 image 512 × 512 × 3 = 786,432 numbers are produced per image, per denoising step

- The diffusion denoiser reads and rewrites all 786k numbers, for each of its ~1000 timesteps, for every image you generate
- Not an amortized cost: Training AND inference both incur it repeatedly
- Pixel-space diffusion works, but at scale it needs a cluster, not a laptop or a single GPU
- **The question:** can we run the exact same diffusion process more economically?
  - YES – run the diffusion process in a low dimensional latent space

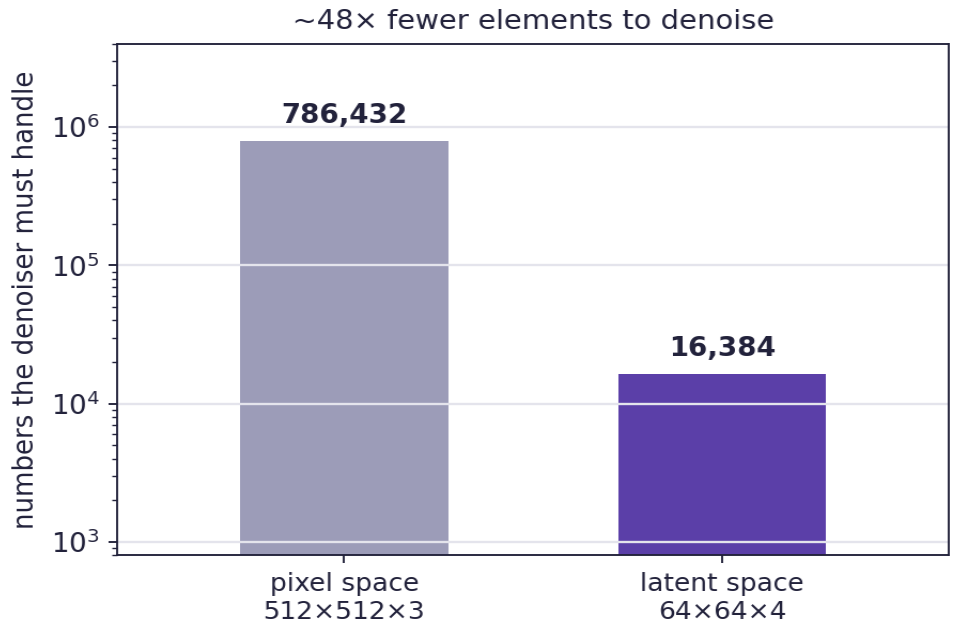
"An oil painting of a space shuttle"

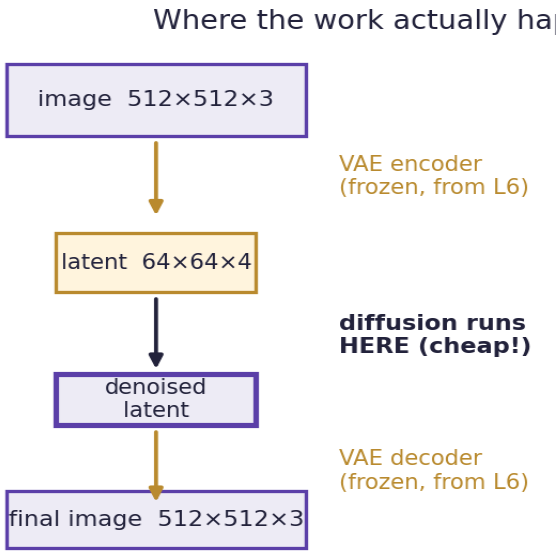


Cost analysis follows Rombach et al. 2022, §1. Denoising objective: Ho et al. 2020 (DDPM).

The fix: diffuse in latent space, not pixel space

Latent diffusion: compress once, denoise cheaply, decode once





- Recall: the VAE encoder already compresses an image into a compact latent representation, trained to keep what matters for reconstruction. We now run the entire diffusion process — forward noise and reverse denoising — inside that latent space, then decode once at the end.
- VAE encoder/decoder trained once, separately, then frozen – VAE handles accurate and efficient encoding of image characteristics
- Inference (new image): just chose a random seed in latent space, then run diffusion denoising, then reconstruct denoised latent to image via VAE decoder
  - cheap and basis of “stable Diffusion” method

Rombach et al. 2022, §3 & Fig. 2 Claude - (Latent Diffusion / Stable Diffusion architecture).

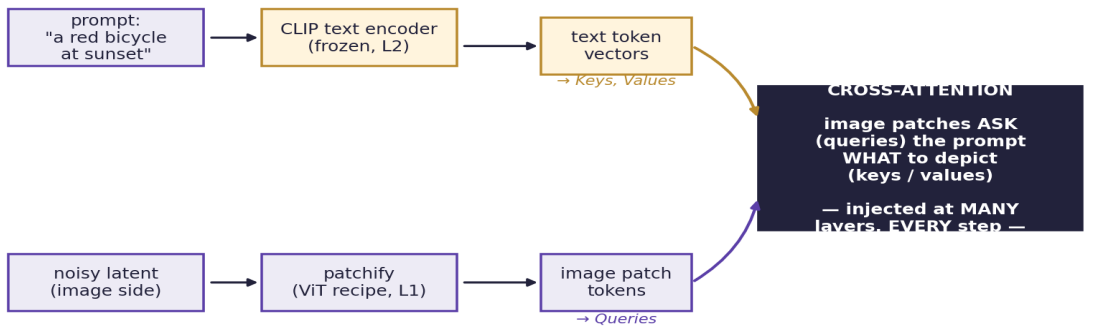
PART II

Text Conditioning

The control problem: how to condition on text?

- Diffusion denoiser network takes (noisy latent, timestep) → predicted noise.
- There is no text anywhere — it can only sample some plausible image, never a specific image type dictated by a prompt
- We need to somehow inject a prompt and ensure it's meaning serves as conditioning at every one of the ~50 denoising steps, not just as a one-time influence at the start.
- **Two sub-problems to solve - we have the tools to understand what is needed already:**
  - Encode the words: use CLIP's text encoder — it already maps prompts into a space aligned with image content (**NOTE**: we use *All* CLIP text tokens – 77 usually - **not** just the [CLS] output as is usually done)
  - Condition the denoiser on them: use attention — with the latent supplying queries, and the text supplying keys and values

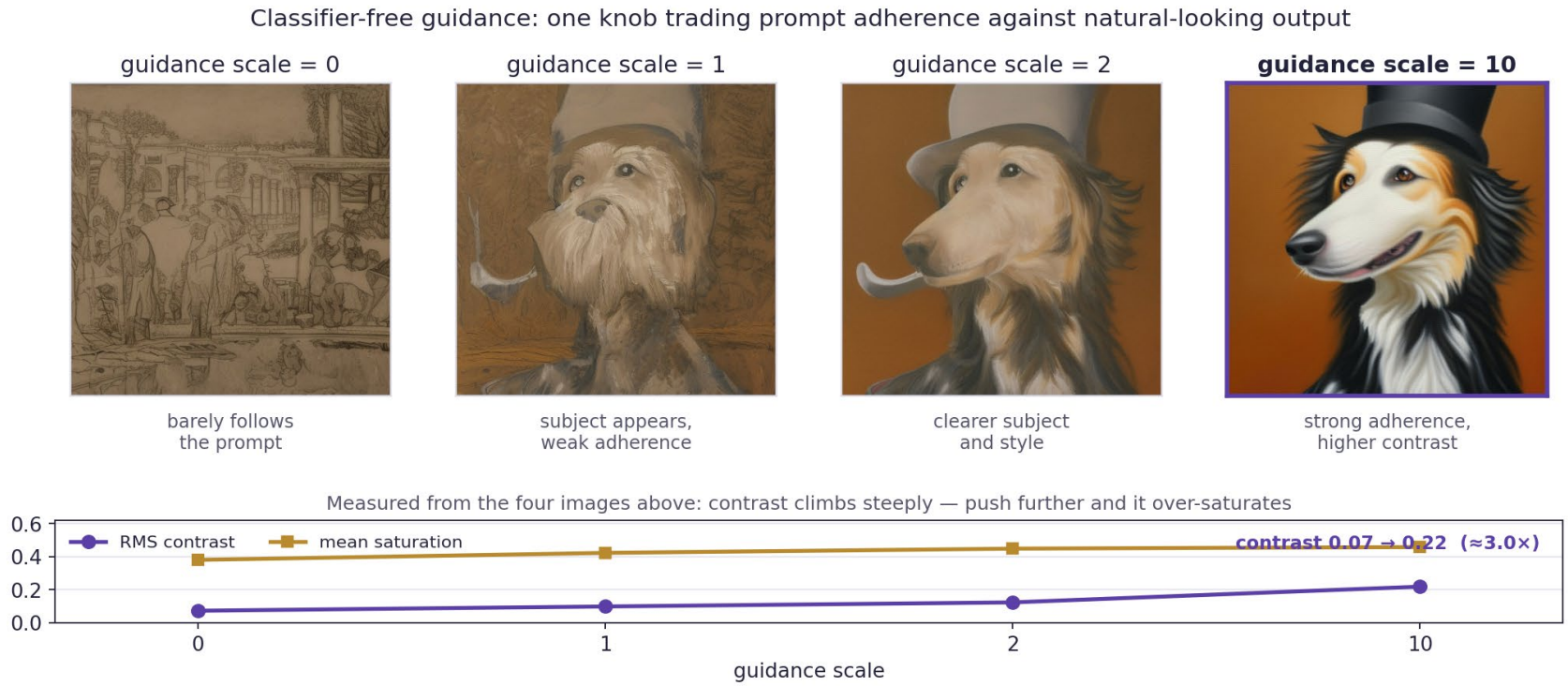
Cross-attention: L1's attention mechanism, repurposed as the steering channel



- Inside the denoiser, cross-attention lets each patch of the noisy latent "ask" the prompt what it should depict — the text supplies the keys and values, the image the queries.
- Injected at many layers in denoising network, at every step, so guidance persists throughout denoising.

Figure: Claude

Classifier-free guidance: the “prompt adherence” knob



- Prompt: “An oil painting of a collie in a top hat” (just because? ;)
- Same model, same prompt, same seed across all four — the only thing changing is the guidance scale.
- Adherence to prompt improves as it rises, but so does contrast:
  - push well past this range and images over-saturate and pick up artifacts.
- Note: guidance scale 0 is unconstrained by prompt – so Claude’s image labelling is a bit off….

Guidance

final noise prediction = (no-prompt prediction) + scale × [(with-prompt) − (no-prompt)]

the bracket is "what the prompt changed"; scale amplifies it

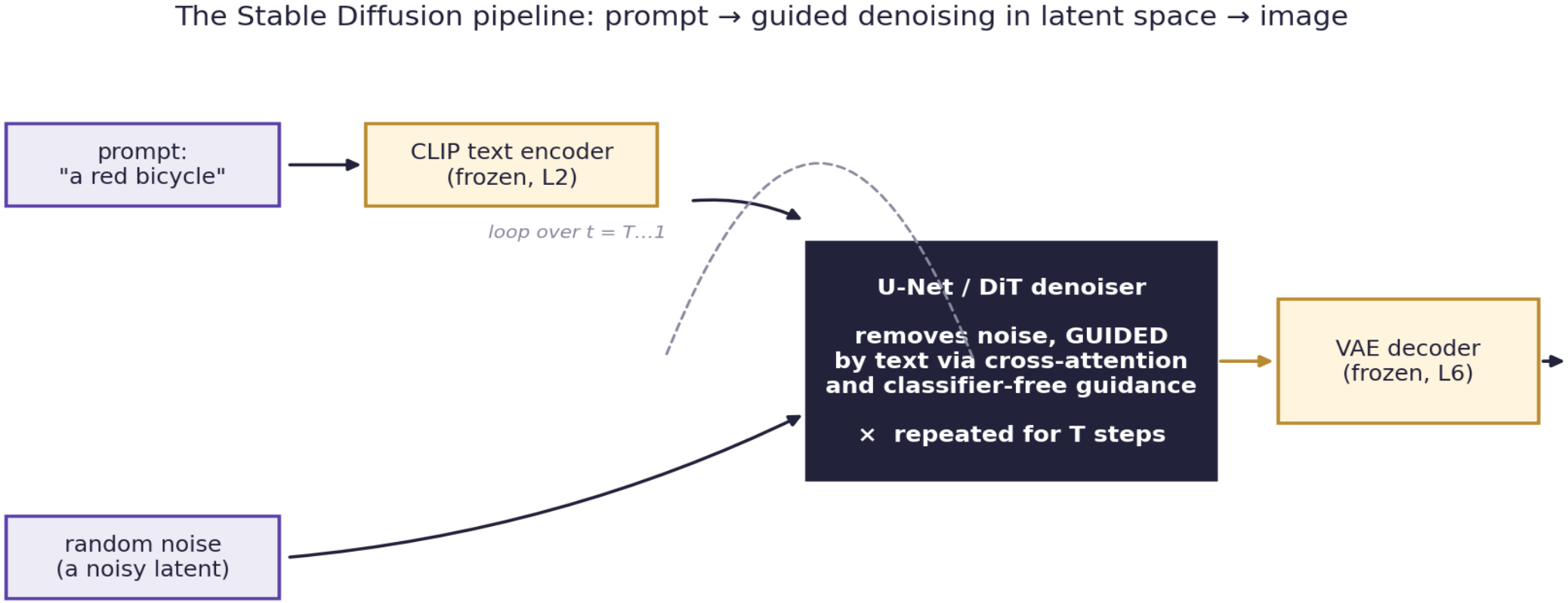
- NOTE: this is a modification of the *noise predicted by the network* (which must then be subtracted from current z during latent denoising to get a 'less noisy latent')
- What the (abstracted) formula above shows**:**
  - The difference (with-prompt) − (no-prompt) is the direction the prompt is pulling the image
  - scale = 1 → ordinary conditioning. scale = 0 → prompt ignored. scale = 7.5 → follow that direction 7.5× as hard
- Why it's called "classifier-free": earlier methods steered generation with a separate image classifier
- this achieves the same steering using only the model's own with/without-prompt predictions — no extra classifier needed
  - The cost: two denoiser passes per step instead of one — guidance roughly doubles generation time.
  - A very common, very worthwhile trade.
  - Training-side note: the model is trained both with and without the prompt (the prompt is randomly dropped during training), so it learns to produce both predictions from training.

Ho & Salimans 2022, arXiv:2207.12598.

PART III

The Full Pipeline

The Stable Diffusion pipeline

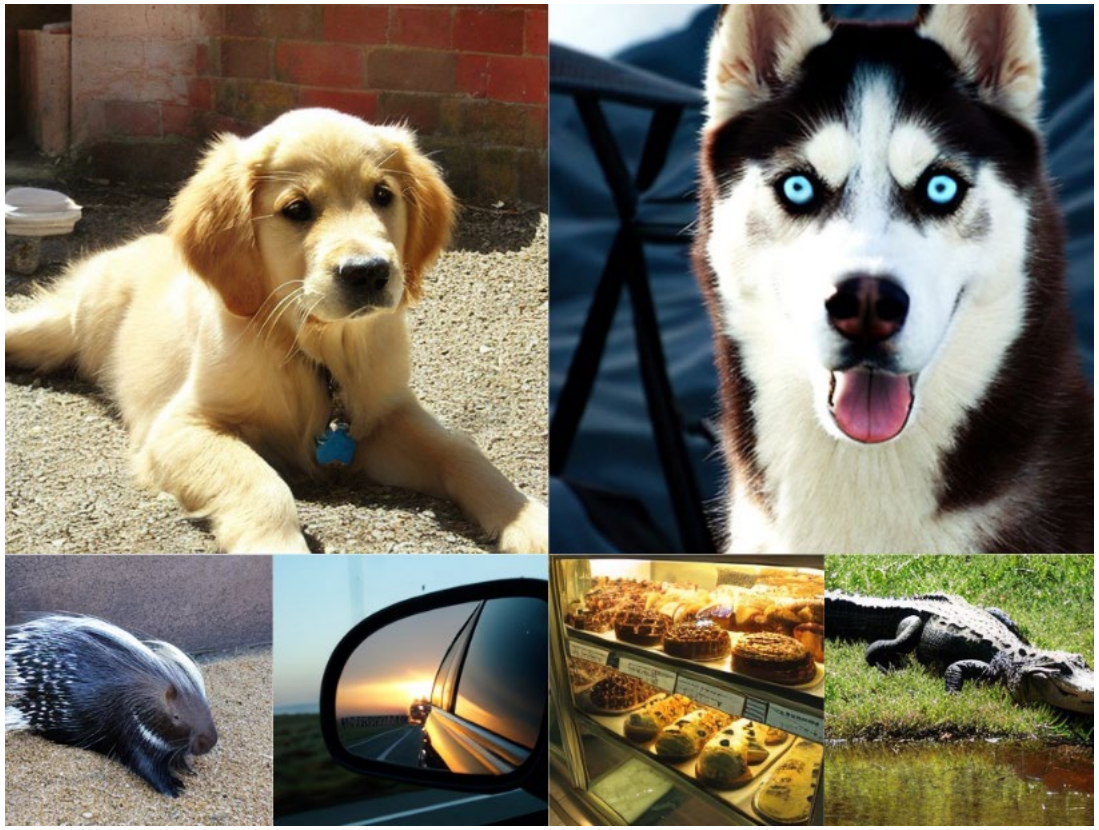


Every block here is an earlier lecture — this IS Stable Diffusion.

Figure: Claude.

More recent upgrade: Diffusion Transformers

- In the pipeline you just saw, the denoiser was a U-Net (a convolutional architecture).
- It works well — but there's a natural alternative: Diffusion Transformer (DiT)
- Replaces the U-Net denoiser with a transformer denoiser
  - The latent ('image') is cut into patches — like a ViT tokenization— and a transformer predicts the noise over those patch tokens
  - Text conditioning still enters by cross-attention (or by conditioning tokens)
  - Guidance remains unchanged
  - Why it matters: DiTs scale better with data and compute, and they're behind most current top-tier image and video generators



Peebles & Xie 2023 (Diffusion Transformers / DiT), arXiv:2212.09748. Image from Peebles & Xie 2023.

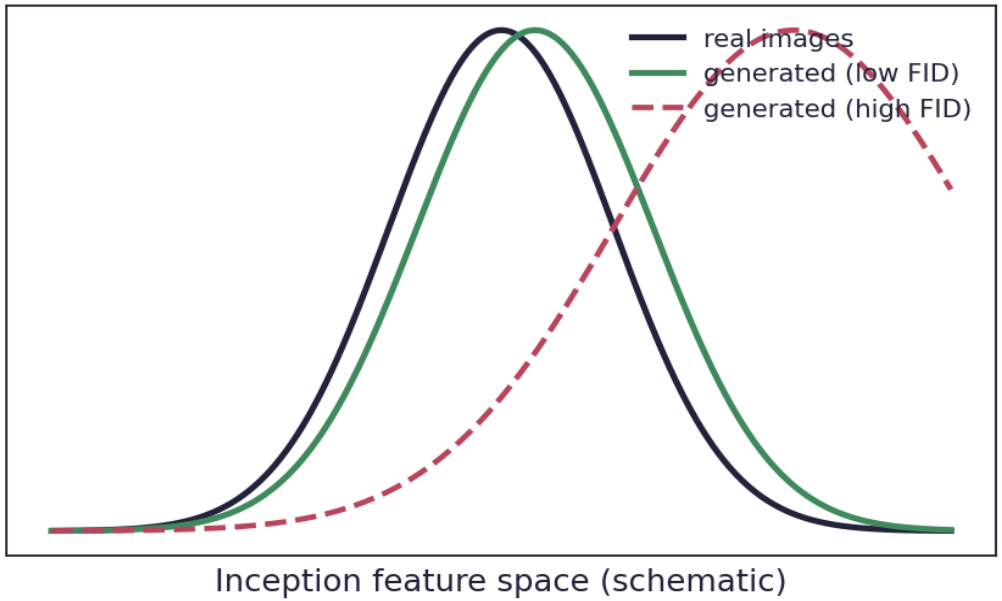
PART IV

Evaluation

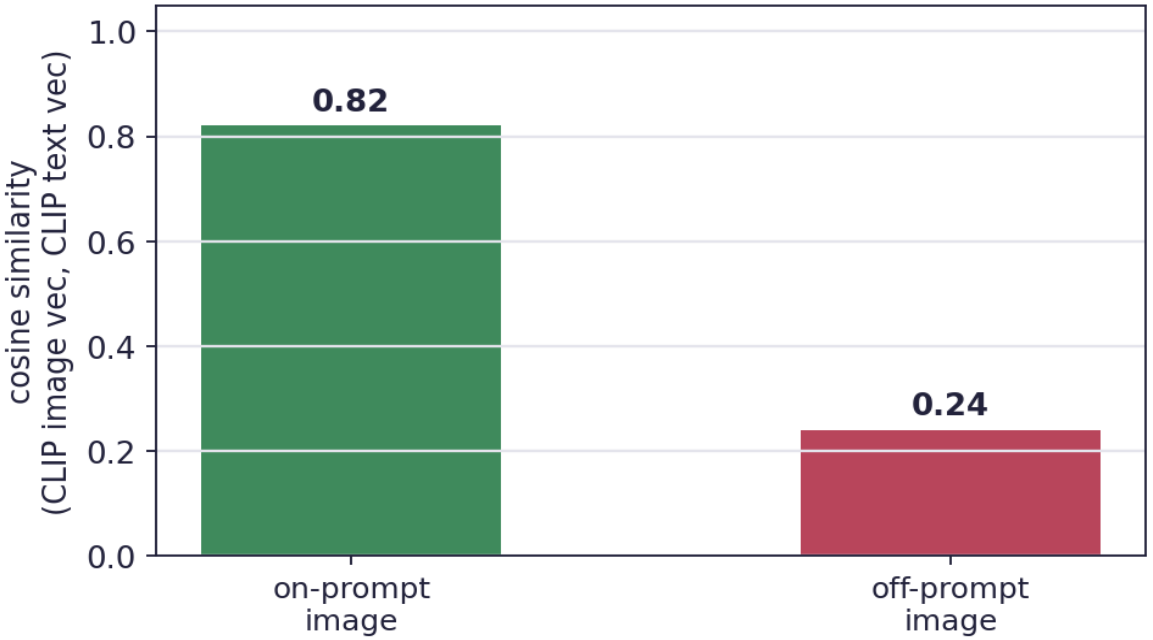
Scoring a generated image: two separate questions

Two questions, two metrics: is it REALISTIC (FID) and does it MATCH the prompt (CLIPScore)?

FID: how close are the two clouds of features? (mean + covariance)



CLIPScore: does the image match THIS prompt? (CLIP cosine, from L2)



- A good generation must be BOTH realistic AND faithful to the prompt — and these can fail independently (a amazing photo of the wrong thing; an on-prompt but ugly mess). So we measure them separately.

Heusel et al. 2017 (FID), arXiv:1706.08500; Hessel et al. 2021 (CLIPScore), arXiv:2104.08718. Figure: Claude.

FID and CLIPscore -

- Fréchet Inception Distance (FID) measures the distance between two feature distributions — of real images and generated images — as seen by a pretrained Inception network
  - Concretely: fit a Gaussian to each set of features (mean + covariance), then measure how far apart those two Gaussians are. Small distance = the generated set “looks like” the real set, (statistically speaking…waves hands…)
- What FID measures: realism and diversity of the generated *set as a whole*. (Lower is better)
  - What FID does not measure: whether any single image matches YOUR prompt — a model can have excellent FID while cheerfully ignoring what you typed
  - Limits: needs a reasonably large sample to estimate the distributions reliably; says nothing about text alignment; sensitive to the specific reference set used
- CLIPScore: embed the generated image and the prompt with CLIP, then take the cosine similarity between the two vectors
  - High cosine similarity → the image and the text sit close together in CLIP’s shared space → the image is on-prompt. It's the same similarity used for zero-shot classification, just pointed at generation instead
- **Why you need BOTH metrics: a model can have good FID while ignoring the prompt entirely (realistic, wrong image), or good CLIPScore while producing a garbled mess that technically contains the right concepts. They catch different failure modes**
  - Neither metric replaces a human looking at the image — they're cheap, automatic proxies for two of the things a human would judge, used at scale during development

Summary

Latent diffusion

Run diffusion in the VAE's compressed latent , not on pixels — ~48× fewer numbers. The VAE handles perception; diffusion handles generation. This is what made image generation affordable.

Text conditioning

Encode the prompt with CLIP (all CLIP tokens used); inject it into every denoising step via cross-attention. Classifier-free guidance is one knob amplifying "what the prompt changed" — trading adherence against quality (SD default ≈ 7.5).

The pipeline = Stable Diffusion

Prompt → CLIP → guided denoising in latent space → VAE decode. Swapping the U-Net denoiser for a transformer gives a DiT — the ViT now used for generation.

Evaluation

FID asks "is it realistic?" (distance between real and generated feature distributions); CLIPScore asks "does it match the prompt?" (CLIP cosine). Need both; humans still judge quality.

Resources

- **References**
  - Rombach et al. (2022). High-Resolution Image Synthesis with Latent Diffusion Models (Stable Diffusion). arXiv:2112.10752
  - Ho & Salimans (2022). Classifier-Free Diffusion Guidance. arXiv:2207.12598 · Peebles & Xie (2023). Scalable Diffusion Models with Transformers (DiT). arXiv:2212.09748
  - Evaluation: Heusel et al. (2017). FID. arXiv:1706.08500 · Hessel et al. (2021). CLIPScore. arXiv:2104.08718
  - Kingma & Welling 2014 (VAE, L6) · Ho et al. 2020 (DDPM, L6) · Radford et al. 2021 (CLIP, L2) · Dosovitskiy et al. 2020 (ViT, L1) · Vaswani et al. 2017 (attention, L1)
- **Code to try:**
  - Hugging Face Diffusion Course, unit 3 (HuggingFace diffusers, Stable Diffusion pipeline) — huggingface.co/learn/diffusion-course